

Data Management 2 (ACSAI)

Gianmaria Romano

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Disclaimer

These notes were taken during the "Data Management and Analysis - Unit 2" course given by Professor Quattrociochi during the second semester of the Academic Year 2024/2025 at the course "Applied Computer Science and Artificial Intelligence" of Sapienza Università di Roma.

Keep in mind, however, that these notes do **not** replace the course material, so it is suggested to check the professor's resources as well.

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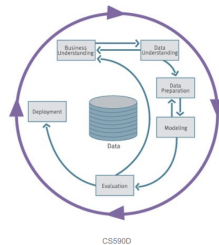
Chapter 1

Complex Systems

1.1 The CRISP-DM process for data mining

Nowadays, the European Community is funding the Cross Industry Standard Process for Data Mining (CRISP-DM) with the aim of finding a standard process that is able to encourage inter-operability and project planning/management.

CRISP-DM: Overview



In particular, the CRISP-DM process starts by understanding the project requirements so that initial data can be selected and cleaned for the data mining phase, at the end of which it will be possible to determine whether the results meet the business objectives, eventually continuing or repeating the data mining step.

1.2 Understanding complex systems

A complex system follows the idea that the collective behaviour of the parts it is composed by leads to the emergence of properties that could be rarely (or never) inferred from the properties of said parts alone.

In particular, complex systems are based on complex networks, which are studied by network science, an interdisciplinary and empirical (data-driven and computational) branch of complex system analysis.

Example: The human brain can be considered as a complex system whose network is represented by the connections between its neurons.

1.3 Social media mining by sentiment analysis

Social media mining is a new network-based strategy to understand/predict society through sentiment analysis, which, on average, has a lower latency and cost while also allowing to cover larger parts of the population, although this method is more prone to phenomena such as polarization or the echo chamber effect.

Generally speaking, sentiment analysis is carried out using domain-specific data and social media data in order to discover eventual trends or correlation, although it should be noticed that correlation typically does not imply causality and that, in some cases, correlation is not a good predictor as it can be spurious.

Part I

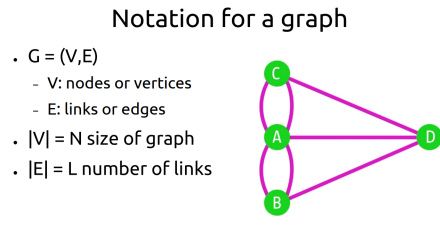
Statistical Tools

Chapter 2

Graph theory

2.1 The geometry of a graph

A (complex) network is normally represented by a graph $G(V, E)$ that is defined using a set of nodes V and a set of links E .



Generally speaking, a graph can be classified as "directed", if it has one-way links, which means that $(u, v) \in E \nRightarrow (v, u) \in E$, or as "undirected", if it has two-way links, which means that $(u, v) \in E \Rightarrow (v, u) \in E$.

2.1.1 Sparse graphs

Theoretically, a graph $G(V, E)$ with N nodes can have up to $L_{\max} = \binom{N}{2}$ links, although, generally speaking, most networks tend to be sparse, meaning that they only feature $L \ll L_{\max}$ connections.

2.1.2 Cliques

A clique in a graph $G(V, E)$ is any (sub-)graph $G'(V', E')$ such that $E' = V' \times V'$. In particular, a complete graph formed by n nodes can be defined as a n -clique.

2.2 The degree of a node

Given a graph $G(V, E)$, a node $i \in V$ is said to have degree k_i if k_i links are incident to it.

Generally speaking, real-life networks can sum up the degree of the nodes by simply considering the average degree, which can be computed as follows:

$$\bar{k} = \frac{2L}{N} = \frac{1}{|V|} \sum_{i=1}^{|V|} k_i$$

N.B.: In the case of a directed graph, it is possible to consider the degree of a node $i \in V$ as $k_i = k_i^{in} + k_i^{out}$, where k_i^{in} denotes the number of incoming links while k_i^{out} denotes the number of outgoing links.

2.2.1 The degree distribution

Given a graph $G(V, E)$ with N nodes, if N_k nodes have degree k , then it is possible to define a probability distribution with weights $p_k = \frac{N_k}{N}$ that allows to derive the average degree of the network through the expected value of the distribution, which can be computed as follows:

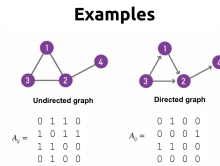
$$\bar{k} = \sum_{k=0}^{\infty} k p_k, \text{ for each possible degree.}$$

Generally speaking, however, most real-life networks follow a linear degree distribution with $\bar{k} = +\infty$, meaning that the degree distribution would not provide useful information about the network.

2.3 Summarising graphs through the adjacency matrix

The adjacency matrix of a graph $G(V, E)$ is a matrix A of size $|V| \times |V|$ such that $A_{ij} = 1$ (or $A_{ij} \in \mathbb{R}^+$ in the case of a weighted graph) if the nodes i and j are connected or $A_{ij} = 0$ otherwise.

Most particularly, assuming $G(V, E)$ is a directed graph, it is possible to compute the in-degree of a node by summing the entries in the corresponding column of the adjacency matrix and the out-degree by summing the entries in the corresponding row of the adjacency matrix.

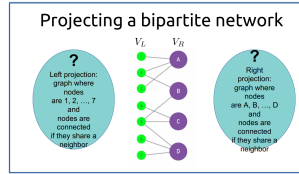


Furthermore, the structure adjacency matrix of a graph can also provide useful information about the graph's structure:

- The graph is undirected if and only if its adjacency matrix is symmetric.
- The graph has self-loops, meaning that a node routes to itself as well, if and only if $\exists i$ such that $A_{ii} \neq 0$.
- The graph is complete, meaning that there always exists a path connecting any two nodes of the graph, if and only if $A_{ij} \neq 0 \forall i \neq j$.

2.4 Bipartite graphs

A bipartite graph is composed by two types of nodes, meaning that it can be represented as a graph $G(V, E)$ where $V = V_L \cup V_R$, with $V_L \cap V_R = \emptyset$, and $E \subseteq V_L \times V_R$.



Generally speaking, a bipartite network can be summarised using an adjacency matrix A of size $|V_L| \times |V_R|$ that can be projected with respect to the left-side nodes using a matrix $A_L = AA^T$ or with respect to the right-side nodes using a matrix $A_R = A^T A$.

2.5 Paths on graphs

A path on a graph $G(V, E)$ is a sequence of edges that connects two nodes. Generally speaking, two nodes $i, j \in V$ are said to belong to the same connected component if there exists a path between them, allowing to state that a graph is connected if and only if it features only one connected component.

2.6 The local clustering coefficient

Given a graph $G(V, E)$, the local clustering coefficient of a node $i \in V$ represents the proportion of connections among its direct neighbours, which can be expressed as follows:

$$C_i = \frac{L_i}{L_{\max}} = \frac{2e_i}{k_i(k_i - 1)}, \text{ with } C_i = 0 \text{ if } k_i \leq 1.$$

Chapter 3

Linear regression

3.1 Modelling a linear regression

Linear regression is a powerful model for establishing a relationship between an explanatory (independent) variable X and a response (dependent) variable y in order to make predictions on future data.

Generally speaking, a linear regression model is built under the assumptions of linearity, meaning that the variables follow a linear relationship, independence of observations and normality and constant variance of the error terms.

In particular, it is possible to consider two main version of this model:

- **Simple regression:** this regression model only involves one independent variable and has equation $y = \beta_0 + \beta_1 X + \varepsilon$, where β_0 represents the y -intercept, β_1 represents the slope of the line and ε represents the error.
- **Multiple regression:** this regression model involves n independent variables $X_1, \dots, X_n$ and has equation $y = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n + \varepsilon$, whose meaning is basically a generalization of the simple regression model.

In any case, the coefficients $\beta_0, \dots, \beta_n$ are estimated by residual analysis, which consists in minimising the square error obtained from the model's predictions.

3.2 The determination coefficient of a linear regression

The determination coefficient of a linear regression is a tool that represents the proportion of variance of the response variable that can be explained through the regression model, meaning that:

$$R^2 = \frac{\sum_{i=1}^N (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \in [0, 1]$$

Chapter 4

Data distributions

4.1 Representing statistical distributions

Generally speaking, it is possible to summarise statistical distributions by using dedicated functions, which represent a continuous normalization of discrete histograms.

4.1.1 Probability mass functions

Assuming a distribution is discrete, meaning that a variable can take a finite (or at least countable) number of possible values, the probability mass function simply provides the probability for each value that the variable can take.

4.1.2 Probability density functions

Assuming a distribution is not discrete and can instead take any possible value within a real-valued set, the probability density function represents the proportion of area under a curve within a given interval:

$$\mathbb{P}(X \in [a, b]) = \int_a^b f(x) \, dx, \text{ with the condition that } \int_{-\infty}^{+\infty} f(x) \, dx = 1$$

N.B.: $f(x)$ must always be a non-negative function.

4.1.3 Cumulative distribution functions

The cumulative distribution function of a continuous distribution denotes the probability that an observation falls below a threshold x , which can be computed in the following way:

$$F(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f(t) \, dt$$

On the other hand, the complementary cumulative distribution function acts as the complement of the cumulative distribution function, meaning that:

$$\overline{F}(x) = \mathbb{P}(X > x) = 1 - \mathbb{P}(X \leq x) = 1 - F(x) = 1 - \int_{-\infty}^x f(t) dt$$

N.B.: $F(x)$ must always be an increasing, non-negative function.

4.2 Random variables and common distributions

A random variable is a function $X : \Omega \rightarrow S_X \subseteq \mathbb{R}$ that measures the probability distribution of the outcomes in a random experiment.

N.B.: Depending on the support S_X , a random variable can be classified as discrete or continuous.

4.2.1 Normal distribution

$X \sim \mathcal{N}(\mu, \sigma^2)$ has a symmetric, bell-shaped distribution centered in μ (which represents mean, median and mode of the distribution) and with spread σ whose probability density function is defined as follows:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \text{ for } x \in \mathbb{R}.$$

Generally speaking, a normal distribution follows an empirical rule, which states that 68% of values falls within one standard deviation from the mean, 95% of the values falls within two standard deviations from the mean and 99.7% of the values falls within three standard deviations from the mean.

Most particularly, the central limit theorem states that, under certain conditions, the behaviour of the mean of a large number of independent and identically distributed variables can be approximated by a normal distribution.

4.2.2 Uniform distribution

$X \sim \text{Uniform}(a, b)$ defines a probability distribution of equally likely outcomes through the following probability density function:

$$f(x) = \frac{1}{b-a}, \text{ for } x \in [a, b].$$

Generally speaking, the uniform distribution is employed in Monte Carlo methods for generating random numbers.

4.2.3 Exponential distribution

$X \sim \text{Exponential}(\lambda)$ models the time passing between Poisson events by means of the following probability density function:

$$f(x) = \lambda e^{-\lambda x}, \text{ for } x \geq 0.$$

Overall, the exponential distribution is characterised by a "memoryless" behaviour as future probabilities do not depend on how much time has passed, meaning that it can be used to define survival functions.

4.2.4 Chi-Squared distribution

$X \sim \chi^2(k)$ is a particular case of the Gamma distribution that is used in the hypothesis test for categorical variables by means of the following probability density function:

$$f(x) = \frac{x^{\frac{k}{2}-1} e^{-\frac{x}{2}}}{2^{\frac{k}{2}} \Gamma(\frac{k}{2})}, \text{ for } x > 0.$$

Generally speaking, this distribution is right skewed, but, by increasing the degrees of freedom, it gradually becomes symmetric.

4.2.5 Student's T distribution

A Student's T distribution estimates deviations of a normal distribution with unknown standard deviation by means of the degrees of freedom.

In fact, it is possible to notice that, as the number of degrees of freedom increases, this distribution can be approximated to a standard normal distribution.

4.3 Covariance and correlation

Given two variables X and Y with means $\bar{x}$ and $\bar{y}$, if these variables are measured on the same N individuals, then their sample covariance provides an idea of the direction of the linear relationship between the variables, meaning that:

$$Cov(X, Y) = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})$$

Most particularly, the Pearson correlation coefficient provides a standardized interpretation of the covariance by setting:

$$\rho_{XY} = \frac{Cov(X, Y)}{s_X s_Y} \in [-1, 1]$$

Overall, $\rho_{XY} \gtrless 0$ denotes a positive/negative relationship, whereas $\rho_{XY} \approx 0$ denotes no linear relationship, although the variables may still have a non-linear relationship.

N.B.: Remember that correlation does not mean causation as two variables might be correlated by pure chance or by another underlying cause, such as another external variable.

4.3.1 The Spearman correlation coefficient

The main issue of the Pearson correlation coefficient lies in the fact that it can only capture linear relationships.

For this reason, the Spearman correlation coefficient provides a more flexible interpretation by computing the correlation coefficient between the ranks of the data points, meaning that:

$$\rho_S = \rho_{\text{rank}(X)\text{rank}(Y)}$$

4.3.2 Covariance and correlation matrices for multivariate data

Generally speaking, when dealing with multivariate data, it can be convenient to use the following data structures:

- **Covariance matrix:** a matrix C such that $C_{ij} = C_{ji} = \text{Cov}(X_i, X_j)$.
- **Correlation matrix:** a matrix R such that $R_{ij} = R_{ji} = \rho_{X_i X_j}$.

Most particularly, it is possible to observe that, by their definition, these matrices are always symmetric.

4.4 The hypothesis test

Generally speaking, a hypothesis test is a statistical procedure that uses empirical evidence from data in order to determine if a null hypothesis about a population parameter is true or if an alternative hypothesis is more reasonable.

4.4.1 The correlation significance test

The correlation significance test aims to determine whether the correlation between two quantitative variables X and Y is statistically significant through the following hypothesis test:

$$\begin{cases} H_0 : X \text{ and } Y \text{ are not significantly correlated} \\ H_1 : X \text{ and } Y \text{ are significantly correlated} \end{cases}$$

Assuming the null hypothesis is true, it is possible to compute the following test statistic representing the sample:

$$t = \frac{\rho_{XY} \sqrt{N-2}}{\sqrt{1-\rho_{XY}^2}} \sim T_{N-2}$$

Lastly, given a significance value α , it is possible to compute a p -value associated to the test statistic to conclude that, if $p < \alpha$, meaning that seeing such a statistic under the assumption of the null hypothesis would be too unlikely, then the data provide strong evidence against the null hypothesis, which is therefore rejected in favour of the alternative hypothesis.

4.4.2 The two-sample t test for comparing populations

The two-sample t test is a statistical test that estimates the significance of the difference in the means between two different populations X and Y (of size N and M respectively), which are assumed to be independently and normally distributed with homogeneous variance.

$$\begin{cases} H_0 : \mu_X = \mu_Y \\ H_1 : \mu_X \neq \mu_Y \end{cases}$$

Assuming the null hypothesis is true, it is possible to compute the following test statistic representing the sample:

$$t = \frac{\bar{x} - \bar{y}}{S_p \sqrt{\frac{1}{N} + \frac{1}{M}}} \sim T_{N+M-2}, \text{ where } S_p = \sqrt{\frac{(N-1)s_X^2 + (M-1)s_Y^2}{N+M-2}}.$$

Lastly, given a significance value α , it is possible to compute a p -value associated to the test statistic to conclude that, if $p < \alpha$, meaning that seeing such a statistic under the assumption of the null hypothesis would be too unlikely, then the data provide strong evidence against the null hypothesis, which is therefore rejected in favour of the alternative hypothesis.

4.4.3 The Chi-Squared test for categorical variables

The Chi-Squared test is a non-parametric test that aims to check whether two categorical variables X and Y (with r and c categories respectively) are associated.

$$\begin{cases} H_0 : X \text{ and } Y \text{ are independent} \\ H_1 : X \text{ and } Y \text{ are related} \end{cases}$$

Assuming the null hypothesis is true, it is possible to start by summarising the observations into a dedicated contingency table in order to compute the expected frequencies in the following way:

$$E_{ij} = \frac{\sum_{n=1}^c O_{in} \sum_{m=1}^r O_{mj}}{N}, \text{ where } N \text{ is the total number of observations.}$$

If the expected frequencies are large enough, with the typical threshold being $E_{ij} \geq 5$, it is possible to compute the following test statistic representing the sample:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi^2((r-1)(c-1))$$

Lastly, given a significance value α , it is possible to compute a p -value associated to the test statistic to conclude that, if $p < \alpha$, meaning that seeing such a statistic under the assumption of the null hypothesis would be too unlikely,

then the data provide strong evidence against the null hypothesis, which is therefore rejected in favour of the alternative hypothesis.

4.4.4 The one-way analysis of variance test for comparing means

The one-way analysis of variance test is a parametric test that aims to compare the means of a quantitative variable X across $k > 2$ categories, each with a count of N_i observations that are assumed to be independently and normally distributed with equal variances.

Generally speaking, the test is set up by decomposing the total variance into the between-group sum of squares, which represents the proportion of inter-group variance, and the within-group sum of squares, which represents the variance of a single group, in the following way:

$$SST = SSB + SSW \Rightarrow \sum_{i=1}^k \sum_{j=1}^{N_i} (x_{ij} - \bar{x})^2 = \sum_{i=1}^k N_i (\bar{x}_i - \bar{x})^2 + \sum_{i=1}^k \sum_{j=1}^{N_i} (x_{ij} - \bar{x}_i)^2$$

In fact, this strategy allows to set up the following hypothesis test:

$$\begin{cases} H_0 : \mu_1 = \mu_2 = \dots = \mu_k \\ H_1 : \text{At least one group has a different mean} \end{cases}$$

Assuming the null hypothesis is true, the test defines the mean square values $MSB = \frac{SSB}{k-1}$ and $MSW = \frac{SSW}{N-k}$ in order to compute the following test statistic representing the sample:

$$F = \frac{MSB}{MSW} \sim F(k-1, N-k)$$

Lastly, given a significance value α , it is possible to compute a p -value associated to the test statistic to conclude that, if $p < \alpha$, meaning that seeing such a statistic under the assumption of the null hypothesis would be too unlikely, then the data provide strong evidence against the null hypothesis, which is therefore rejected in favour of the alternative hypothesis.

Most particularly, whenever the null hypothesis is rejected, it is possible to identify which group has a different mean by pairwise comparisons using the Tukey Honest Significant Difference test.

Chapter 5

Data pre-processing

5.1 Data preparation for quantitative variables

Data pre-processing is a procedure that consists in "preparing" a quantitative variable X in order to provide algorithmic compatibility and informative patterns for data comparisons.

5.1.1 Standardization of values

Standardization consists in computing how many standard deviations an observation lies from the mean by computing the corresponding Z -score:

$$z_i = \frac{x_i - \bar{x}}{s_x}$$

In particular, the empirical rule of the normal distribution allows to state that any observation such that $|z_i| > 3$ can be considered as an outlier.

5.1.2 Normalizing values by scaling

Useful for range-sensitive data, scaling consists in setting all the observed data in the interval $[0, 1]$ in the following way:

$$x'_i = \frac{x_i - \min\{X\}}{\max\{X\} - \min\{X\}}$$

5.1.3 Data transformations for skewed distributions

When dealing with skewed distributions, it can be convenient to apply a geometric transformation, such as the logarithmic or the square root distributions, or to consider a boxcox transformation, which tries to approximate the distribution to a normal curve.

5.1.4 Binning quantitative variables into categorical variables

Binning is a procedure that allows to transform a continuous variable into a categorical variable by creating some thresholds, which can be set manually or through quantiles, in order to simplify the interpretation of the distribution.

5.2 Creating new variables by feature engineering

Feature engineering consists in creating new variables for a dataset by mixing its existing variables according to domain-informed logic or known information about the dataset.

5.3 One-hot encoding for categorical variables

Given a categorical variable with k possible categories, dummy encoding consists in building $(k - 1)$ binary indicators that allow to specify whether an observation belongs to a certain category or not.

5.4 Handling missing values

Generally speaking, it is possible to handle missing values either by deletion, which is used when the values are missing at random, or by imputation, which focuses on the mean/median or on the regression model.

Part II

Data Mining Strategies

Chapter 6

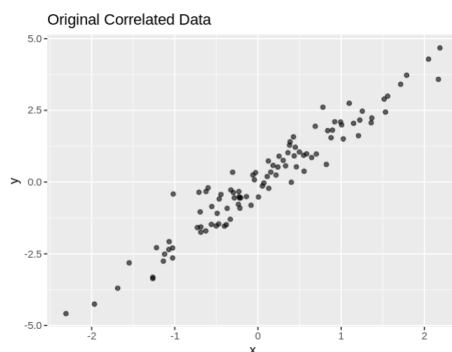
Principal component analysis

6.1 Understanding principal component analysis

Principal component analysis is a statistical technique that allows to reduce the dimensionality of a dataset while keeping as much variance as possible by focusing on the more informative features.

Generally speaking, given a dataset X representing a sample of N points, each with p features, it is possible to determine the most informative attributes by computing the covariance matrix of the dataset in the following way:

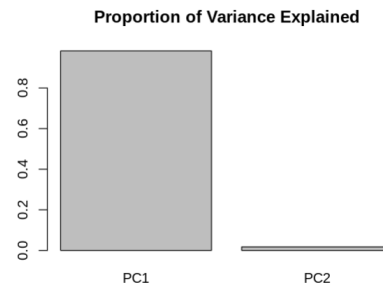
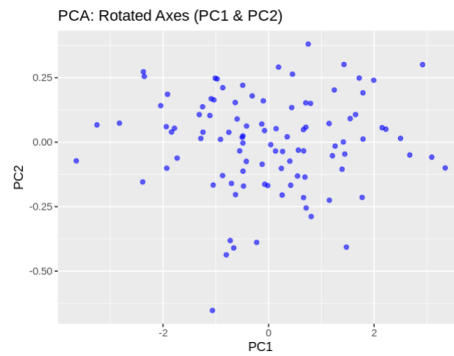
$$\Sigma = \frac{1}{N-1} \bar{X}^T \bar{X}, \text{ where } \bar{X} = X - \mu \text{ represents the centered dataset.}$$



In fact, the covariance matrix allows to create a new set of variables, known as the "principal components", which will be uncorrelated and decreasingly sorted

by the proportion of variance they can explain, which can be quantified as follows:

$$PVE_k = \frac{\lambda_k}{\sum_{j=1}^p \lambda_j}, \text{ where } \lambda_1, \dots, \lambda_p \text{ are the eigenvalues of } \Sigma.$$



Overall, principal component analysis comes in handy to summarise data with highly correlated features or to spot patterns/anomalies through clustering.

Chapter 7

Segmentation

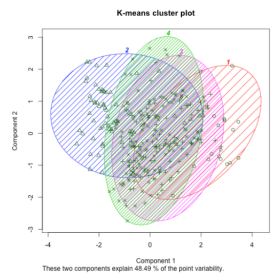
7.1 Defining segmentation

Segmentation consists in partitioning a dataset into clusters in such a way that data points in the same cluster will be more similar to each other compared to other data points.

Generally speaking, it is also possible to visualize segmentation using a dendrogram representing the hierarchical clustering of the dataset.

7.2 Clustering by K -means

Clustering is an unsupervised learning model that partitions the dataset in K distinct clusters in order to spot patterns or anomalies within the data points. Generally speaking clustering is carried out using the K -means algorithm, which, after randomly initializing K centroids from the dataset, repeats assignment, which consists in mapping each point to the nearest centroid, and update, which consists in recalculating the centroid positions after the assignment step, until convergence, which is achieved when the centroids do not change within some error term.



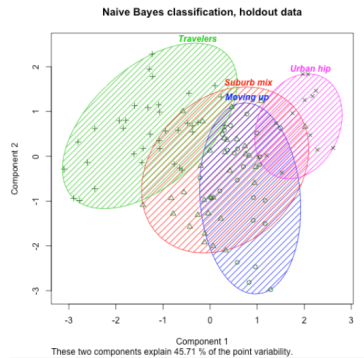
Overall, this algorithm is fast and scalable, but it is only locally optimal because the solution heavily depends on the initialization step.

7.3 Classification by Naive Bayes models

Classification is a supervised learning model, which, based on a hypothesis that was formulated on previous labellings, tries to predict the labels of new data points.

Generally speaking, classification is carried out through Naive Bayes generative models, which revolve around posterior probabilities that are computed through Bayes' theorem as follows:

$$P(C|\mathbf{x}) = \frac{\mathbb{P}(\mathbf{x}|C)\mathbb{P}(C)}{\mathbb{P}(\mathbf{x})}, \text{ for any label } C.$$



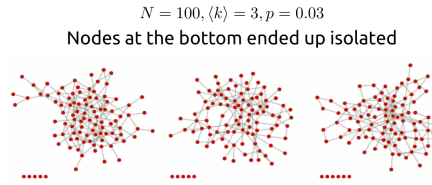
Overall, supervised learning models are often handled using random forests, which, in order to avoid overfitting the dataset, build and merge more decision trees in order to provide more balanced predictions.

Chapter 8

Network science

8.1 Formalizing complex networks through random networks

The Erdős-Rényi model builds a random network $G(N, p)$ by repeatedly performing a Bernoulli trial with success probability p in order to determine whether there should exist a link between two nodes.



Generally speaking, it is possible to study a random network using its degree distribution, which can be defined through the following probability mass function:

$$p_k = \binom{N-1}{k} p^k (1-p)^{N-1-k}, \text{ for every node in the network.}$$

Most particularly, the degree distribution of a random network allows to derive the average degree of a node starting from the expected number of links:

$$\text{Given } \bar{L} = pL_{\max} = p \frac{N(N-1)}{2}, \text{ let } \bar{k} = \frac{2\bar{L}}{N} = p(N-1).$$

Overall, however, the Erdős-Rényi is considered a poor representation of the degree distribution of a network as it often approximates the degree distribution to a Poisson distribution, although this rarely happens in real networks.

8.1.1 Using random networks to study network connectivity

Generally speaking, the degree distribution of a random network also allows to study its connectivity and the corresponding internal changes.

In fact, assuming the network is super-critical, meaning that $\bar{k} > 1$, it is possible to estimate the number of nodes within some distance d of a given node in the following way:

$$N(d) = \sum_{j=0}^d \bar{k}^j = \frac{\bar{k}^{d+1} - 1}{\bar{k} - 1}$$

Overall, this approach makes the Erdős-Rényi model good for estimating path lengths and distances, although it typically provides a poor estimate of the clustering coefficient distribution.

8.2 Understanding bipartite networks

A bipartite network is characterised by two disjoint sets of nodes V_L and V_R , meaning that it can be represented as a graph $G(V, E)$ where $V = V_L \cup V_R$, with $V_L \cap V_R = \emptyset$, and $E \subseteq V_L \times V_R$.

Example: It is possible to use a bipartite network in order to study users' interactions with YouTube videos by considering user and video nodes so that an edge will represent an interaction between a user and a video.

8.2.1 Performing one-mode projection on bipartite networks

A bipartite network is characterised by a biadjacency matrix of size $|V_L| \times |V_R|$ that allows to summarise its connections.

This matrix is generally used to perform one-mode projection, which consists in focusing on the eventual connections between nodes of the same type through one of the following matrix projections:

$$\begin{cases} A_L = AA^T & \text{for the left projection.} \\ A_R = A^T A & \text{for the right projection.} \end{cases}$$

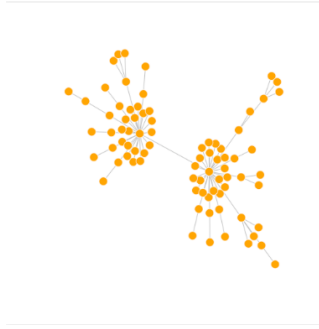
8.3 Graph layout strategies

A graph layout algorithm aims to provide a clear structural insight of a network while also taking its features into consideration.

8.3.1 The Fruchterman-Reingold layout algorithm

The Fruchterman-Reingold algorithm simulates a physical system in which the nodes can experience repulsive and/or attractive forces.

In fact, the algorithm, after randomly initializing the positions of the nodes, repeatedly computes each node's net force displacement either until stabilization, which is achieved when the force changes are not significant anymore, or when a maximum number of iterations is reached.



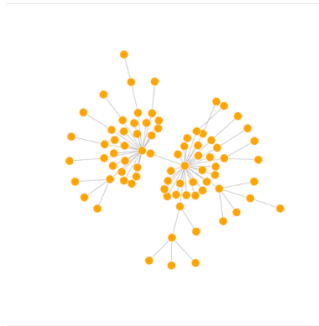
Generally speaking, the Fruchterman-Reingold algorithm is considered to be a good layout tool for general-purpose visualizations of middle-sized networks.

8.3.2 The Kamada-Kawai layout algorithm

The Kamada-Kawai algorithm tries to convey graph-theoretical distances using the following energy function:

$$E = \sum_{u \neq v} \frac{1}{2} k_{uv} (\|p_u - p_v\| - \text{dist}(u, v))^2$$

In fact, the algorithm, after computing the theoretical distance of the initial network (which is typically random or circular), aims to minimise E until convergence or when a maximum number of iterations is reached.



Overall, the Kamada-Kawai algorithm provides a reliable visualization of the network, although it tends to be computationally expensive.

8.4 Network analysis metrics

Generally speaking, a network metric strategy provides tools for analysing network properties.

8.4.1 The degree distribution of nodes

Given a network $G(V, E)$, the degree of a node represents the number of links that are incident on it, which, in the case of a directed network, can be decomposed into an in-degree, which denotes the number of incoming links, and an out-degree, which denotes the number of outgoing links.

Generally speaking, the degree distribution of a network can come in handy to define the reciprocity of a directed network as the proportion of bidirectional connections within the graph.

8.4.2 Performing dyad census on connected nodes

When dealing with directed networks, it is possible to perform a dyad census on two different nodes in order to determine the proportion of mutual relationships, which are given by bidirectional connections, asymmetric relationships, which are given by one-way connections, or null relationships, which concern disconnected couples.

8.4.3 Applying clustering metrics by transitivity

Given a network $G(V, E)$, transitivity consists in performing clustering analysis in order to spot eventual interconnected groups within the network.

Generally speaking, given a node $i \in V$ with degree k_i , if there exist e_i connections between its direct neighbours, it is possible to define the local clustering coefficient of the node as follows:

$$C_i = \begin{cases} \frac{2e_i}{k_i(k_i-1)} & \text{if } k_i > 1 \\ 0 & \text{if } k_i \leq 1 \end{cases}$$

Most particularly, this procedure also allows to define the global clustering coefficient of the network as the average of all clustering coefficients, meaning that:

$$C = \frac{1}{|V|} \sum_{i=1}^{|V|} C_i$$

8.4.4 Applying centrality metrics

A centrality metric is used to understand the accessibility and/or influence of a node within a (connected) network.

Generally speaking, it is possible to measure node centrality through one of the following metrics:

- **Closeness centrality:** the influence of a node is defined in terms of how quickly it can reach the other nodes, resulting in:

$$C(u) = \frac{1}{\sum_{v \neq u} d(u, v)}, \text{ for every shortest path from/to } u.$$

- **Betweenness centrality:** the influence of a node is defined in terms of the proportion of (shortest) paths passing through it, meaning that:

$$C_B(u) = \sum_{s \neq u \neq t} \frac{\sigma_{st}(u)}{\sigma_{st}}, \text{ for all shortest paths between } s \text{ and } t.$$

- **Eigenvector centrality:** the influence of a node is defined in terms of both the quantity, which considers the number of connections the node is able to form, and the quality, which instead considers how strong the node's connections are.

In fact, by solving an eigenvalue problem on the adjacency matrix of the network, it is possible to compute the centrality eigenvector of the network as follows:

$$x_i = \frac{1}{\lambda} \sum_j A_{ij} x_j, \text{ where } \lambda \text{ is the largest eigenvalue of } A.$$

Therefore, it is possible to observe that influential nodes must be connected to other influential nodes.

8.5 Community detection algorithms

Given a network $G(V, E)$, a community is a sub-graph $G'(V', E')$ that is complete, meaning that $V' \subseteq V$ but $E' = V' \times V'$.

In particular, it is possible to assess how much a network can be partitioned into densely connected communities by means of the modularity metric, which, for a graph $G(V, E)$, is defined as follows:

$$Q = \frac{1}{2|E|} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2|E|} \right) \delta(c_i, c_j), \quad \delta(c_i, c_j) = \begin{cases} 1 & \text{if } i, j \text{ are in the same clique} \\ 0 & \text{otherwise} \end{cases}$$

8.5.1 The cluster edge betweenness algorithm

This algorithm tries to detect communities by iteratively removing the edges with the highest betweenness centrality in order to measure their influence over the information flow, making this strategy useful for finding cohesive groups.

8.5.2 The label propagation algorithm

This algorithm exploits local information exchange as it initializes the nodes of the network to have unique labels in order to iteratively update each node's label through a "majority vote" on the node's neighbours, eventually choosing at random in the event of a tie.

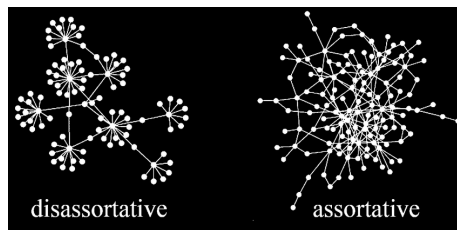
Generally speaking, the label propagation algorithm comes in handy to create communities by observing that the procedure tends to converge to a "local consensus" in the labelling.

8.5.3 Fast greedy optimization

This algorithm greedily tries to optimize the modularity of the network to ensure that community partitioning will be as efficient as possible.

8.6 Network assortativity

Homophily represents the tendency of a node to associate to similar nodes, leading to a "feedback cycle" in which a node chooses similar nodes and gets influenced by them.



Generally speaking, the most common associated metric is assortativity, which defines homophily by node degree.

Chapter 9

Sentiment analysis

9.1 Understanding the main goals of sentiment analysis

Sentiment analysis deals with understanding the polarity of a subjective state that represents the author's sentiment about a target topic.

However, since there is no objective verification for sentiments due to language inconsistencies/shades, it is generally more convenient to start by finding some "ground truth" in order to derive cleaner metrics that also take perspective and authority into consideration.

In particular, the "arousal" of a sentiment represents the strength of an opinion, which can come in handy for making inference about the emotions of the author.

Robert
Plutchik's
Wheel



9.2 Building a pipeline for sentiment analysis

Generally speaking, handling complex datasets for sentiment analysis requires thinking about a general structure (and abstraction) of the pipeline strategy,

which should be transparent, meaning that it is easy to use, reusable, meaning that the same pipeline can be used for different projects, and composable, meaning that the chosen pipeline should be able to analyse large quantities of data.

A sentiment analysis pipeline typically starts from loading the raw data, which are then preprocessed to be suitable for the analysis.

In fact, it is possible to use the preprocessed data in order to perform sentiment extraction through the NRC Sentiment and Emotion lexicon so to analyse the eventual patterns between user activity and expressed sentiment.

Lastly, it is possible to plot a bipartite post-user network in order to analyse the patterns through homophily testing, which can come in handy to understand the sentiment communities.

Appendix A

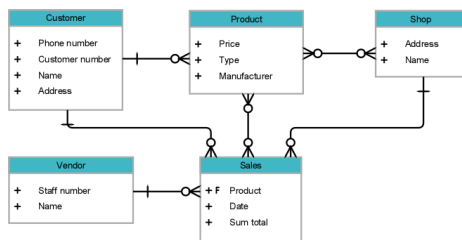
Representing data in SQL

A.1 Understanding the SQL programming language

The SQL programming language provides tools for database managements systems by summarising relational schemas into tables whose rows represent tuples/records (which can be uniquely identified through the key attributes of the schema) and whose columns represent the attributes.

Most particularly, it is possible to design a database management system by means of the entity-relationship model, which provides an efficient visualization of inter-relational systems.

Example: An entity-relationship model can be used to design a purchase database that provides information about customers, products and vendors.



A.2 Performing basic SQL queries

Generally speaking, a query creates new relational schemas by manipulating input relational schemas through dedicated operations.

The simplest SQL query is selection, which consists in selecting tuples that satisfy a certain condition through the **SELECT (*) FROM *schema* WHERE *condition*** command.

In addition, by specifying some *attributes* in place of (*), it is possible to project the selected tuples over the chosen attributes.

A.2.1 Additional SQL tasks

When performing a SQL query, it is also possible to remove duplicates from a schema, which can be done by specifying the **DISTINCT** keyword, or increasingly sort records with respect to a category, which can be done through the **ORDER BY** *category* command.

A.3 Joining more tables

The **JOIN** operator allows to combine more tables through the matching common attributes, eventually specifying a **LEFT/RIGHT/FULL** keyword to include some non-matching components as well.

Furthermore, it is also possible to join a table by itself through the "*attribute AS alias*" syntax within the operation's scope.

A.4 Data aggregation

Data aggregation consists in performing some operations, such as **SUM**, **COUNT**, **MIN**, **MAX** or **AVG**, on a specific attribute.

Most particularly, data aggregation also allows to perform additional tasks, such as grouping, which consists in grouping data by an attributed through the **GROUP BY** *attribute* command, and/or conditioning, which instead consists in setting constraints over groups through the **HAVING** *condition* command.

A.5 Data manipulation

It is possible to modify the records of relational schema by performing the following operations:

- **Insertion:** add tuples using the **INSERT INTO** *schema* **VALUES** *values* command.
- **Deletion:** remove tuples using the **DELETE FROM** *schema* **WHERE** *condition* command.
- **Update:** modify tuples using the **UPDATE** *schema* **SET** *new* **WHERE** *condition* command.